Swarm Intelligence and Target Search

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A "treasure hunt" can be a fun and engaging opportunity to test the capabilities of a swarm of robots in a simulated environment. The multiagent system proposed in this project is designed to search for a target in a 2D environment using a combination of swarm intelligence (? ?), Voronoi cells (? ?), Lévy flight (? ?) and sensor fusion techniques. The main objective of the project is to evaluate the performance of the proposed multiagent system in terms of search efficiency, convergence time, and robustness to noise and uncertainties in the environment. The simulation will be implemented in Python, and the results will be analyzed to draw conclusions about the effectiveness of the proposed approach in solving the treasure hunt problem.

Introduction

This project presents a control framework for a swarm of mobile robots that finds, estimates and encircles a target. The system combines a lidar, for obstacle detection, an Extended Kalman Filter (EKF) for state estimation and linear consensus for coordination. The treasure is supposed to be a point object emitting a weak radio signal, the detection is performed by a simple thresholding of the received signal strength. The robots are able to coordinate their movements using a linear consensus algorithm, allowing convergence to the estimated position of the target while avoiding obstacles. An iteration of the consensus algorithm enhances the precision of the target estimate. The performance of the proposed control framework is evaluated through simulations in a 2D environment, where the robots are tasked with finding and aligning around a target while navigating through a cluttered environment. The results show that the proposed approach is effective in enabling the swarm of robots to find and align around the target, even in the presence of noise and uncertainties in the environment.

The task

The task is defined, in a 2D environment, as the search of a target, supposed to be a point object. The issue addresses the problem of distributed cooperative control for a group of mobile robots, which are able to act autonomously. The project allows for the implementation of a control framework combining a Voronoi-Lloyd partitional scheme, an Extended Kalman Filter and a linear consensus algorithm. The EKF allows for the sensor fusion of the odometry and the realsense position measurements. The swarm is tasked with exploring, avoiding obstacles, finding the target and aligning around it.

Structure and workflow. The simulation is performed in a 2D environment filled with obstacles emulating a cluttered room. A finite state machine is implemented to control the behavior of the robots. which are listed as follows:



Fig. 1. Placeholder image of Iris with a long example caption to show justification setting.

[noitemsep]Searching: the robots explore the environment using Levy flight while avoiding other robots and obstacles. Inspecting: the robots detects an obstacle and checks if it is the target by getting closer and measuring the signal strength. Treasure found: once the target is found, all agents converge to the estimated position and further refine the estimate by iterating the consensus algorithm.

Just for kicks here's a citation (**?**). And a reference to a supplement Sup. Note 1. And Something about something. Figure 1 shows an example of how to insert a column-wide figure. To insert a figure wider than one column, please use the `\begin{figure*}...\end{figure*}` environment. Figures wider than one column should be sized to 11.4 cm or 17.8 cm wide. Use `\begin{SCfigure*}...\end{SCfigure*}` for a wide figure with side captions.

Conclusions

blablaba 1 Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like "Huardest gefburn"? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the

alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

Blabla. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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ACKNOWLEDGEMENTS

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Bibliography

Word Counts

This section is *not* included in the word count.

Notes on Nature Methods Brief Communication.

- Abstract: 3 sentences, 70 words.
- Main text: 3 pages, 2 figures, 1000-1500 words, more figures possible if under 3 pages

Statistics on word count.

```
File: Article.tex
Encoding: utf8
Sum count: 597
Words in text: 567
Words in headers: 17
Words outside text (captions, etc.): 13
Number of headers: 10
Number of floats/tables/figures: 1
Number of math inlines: 0
Number of math displayed: 0
Subcounts:
  text+headers+captions (#headers/#floats/#inlines/#displayed)
  16+5+0 (1/0/0/0) _top_
  116+1+0 (1/0/0/0) Abstract
  171+1+0 (1/0/0/0) Section: Introduction
  95+2+0 (1/0/0/0) Section: The task
  156+3+13 (1/1/0/0) Subsection: Structure and workflow
  1+1+0 (1/0/0/0) Section: Conclusions
  0+1+0 (1/0/0/0) Subsection: Blabla
  1+1+0 (1/0/0/0) Section: Conclusions
  11+2+0 (2/0/0/0) Section: Bibliography
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Supplementary Note 1: Something about something

Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

This is the second paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

And after the second paragraph follows the third paragraph. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

After this fourth paragraph, we start a new paragraph sequence. Hello, here is some text without a meaning. This text should show what a printed text will look like at this place. If you read this text, you will get no information. Really? Is there no information? Is there a difference between this text and some nonsense like “Huardest gefburn”? Kjift – not at all! A blind text like this gives you information about the selected font, how the letters are written and an impression of the look. This text should contain all letters of the alphabet and it should be written in of the original language. There is no need for special content, but the length of words should match the language.

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